TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2008

MATHEMATICS 9740/01

Higher 2

Paper 1

Tuesday **20 August 2008** **3 hours**

Additional materials: Answer paper
Cover Page
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in, including the Cover Page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Do not write anything on the List of Formulae (MF15).

Answer **all** the questions.
Begin each answer on a **fresh page** of paper.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are required of the need for clear presentation in your answers.

The number of marks is given in brackets **[ ]** at the end of each question or part question. At the end of the examination, place the completed cover page on top of your answer scripts and fasten all your work securely together with the string provided.

For Candidate's Use	For Examiner's Use
Question Number	**Marks Obtained**
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
Total	
Calculator Model:	

This question paper consists of 5 printed pages including this page.

1 A traveller just returned from Germany, France and Spain. The amount that he spent each day on housing, food and incidental expenses in each country are shown in the table below

Country	**Housing**	**Food**	**Incidental Expenses**
Germany	30	30	10
France	20	30	10
Spain	20	20	10

The traveller's records of the trip indicate a total of $420 spent for housing, $440 for food and $180 for incidental expenses. Calculate the number of days the traveller spent in each country.

He did his account again and the amount spent on food is $400. Is this record correct? Why? **[5]**

2 A curve has parametric equations $x = \theta - \sin\theta$ and $y = 1 - \cos\theta$.
Without the use of graphic calculator, show that the gradient of the tangent at the point where $\theta = \dfrac{\pi}{2}$ is 1. Hence find the equation of normal at the same point. **[5]**

3 Given that y satisfies the differential equation $\dfrac{dy}{dx} = (x+y)^2$, and $y = 1$ at $x = 0$.

(i) By successive differentiation of this result, show that

$$\frac{d^3y}{dx^3} = 2\left[(x+y)\left(\frac{d^2y}{dx^2}\right) + \left(1 + \frac{dy}{dx}\right)^2\right].$$ **[3]**

(ii) Express y as a series in ascending powers of x up to and including the term in x^3. **[3]**

4 Show that $x^2 - x + 1 > 0$ for all real x. Hence find the range of values of a if the inequality $\dfrac{x^2 + ax - 2}{x^2 - x + 1} < 2$ is satisfied for all real x. **[6]**

5 Find $\dfrac{d}{dx}\left(e^{\sin^{-1}(x^2)}\right)$ and hence find the exact value of $\displaystyle\int_0^1 \frac{x\cos^{-1}(x^2)}{\sqrt{1-x^4}}\left(e^{\sin^{-1}(x^2)}\right)dx$. **[6]**

6 Given that $z_1 = 1 - 2i$ is a root of the quadratic equation $z^2 + az + b = 0$ where a and b are real, find the values of a and b. Mark on an Argand diagram P_1 and P_2, the points representing the roots z_1 and z_2 of the quadratic equation $z^2 + az + b = 0$. A third point P_3 is on the real axis such that P_1, P_2 and P_3 form a triangle which encloses the origin with an area of 8 square units. Find z_3 represented by the point P_3, hence form a cubic equation having roots z_1, z_2 and z_3. **[7]**

7 **(i)** Show that $\dfrac{1}{2^r(r)}-\dfrac{1}{2^{r+1}(r+1)}=\dfrac{r+2}{2^{r+1}r(r+1)}$. **[1]**

(ii) Find $\displaystyle\sum_{r=1}^{n}\frac{r+2}{2^{r+1}r(r+1)}$ and evaluate $\displaystyle\sum_{r=5}^{\infty}\frac{r+2}{2^{r+1}r(r+1)}$, leaving your answer in the form of $\dfrac{p}{q}$ where p and q are real. **[7]**

8 A sequence of real numbers $x_1, x_2, x_3, \ldots$ satisfies the recurrence relation $x_{n+1}=\dfrac{4x_n}{3+x_n^{\ 2}}$ for $n \geq 1$.

(i) Prove algebraically that, if the sequence converges, then it converges to either -1, 0 or 1. **[2]**

(ii) Use a calculator to determine the behaviour of the sequence for the case $x_1 = 0.5$. **[2]**

(iii) Express $x_{n+1}-x_n$ in the form $\dfrac{P(x_n)}{3+x_n^{\ 2}}$, where $P(x)$ is a polynomial. Show that $P(x)<0$ if $-1<x<0$ or $x>1$. Hence or otherwise deduce that

$x_{n+1}<x_n$ if $-1<x_n<0$ or $x_n>1$

$x_{n+1}>x_n$ if $0<x_n<1$ or $x_n<-1$ **[3]**

(iv) Given that function $\mathrm{f}(x)=\dfrac{4x}{3+x^2}$ is increasing over the interval $-\sqrt{3}<x<\sqrt{3}$, explain briefly how this fact and the results in part **(iii)** relate to the behaviour determined in part **(ii)**. **[2]**

9 The plane Π_1 has equation $5x-3y-4z=2$ and Π_2 has equation $2x+y+5z=15$.

(a) Find the acute angle, correct to the nearest degree, between the planes. **[3]**

(b) Find the equation of the line passing through A with position vector $\mathbf{i}-2\mathbf{j}-3\mathbf{k}$ and parallel to the line of intersection of Π_1 and Π_2. **[3]**

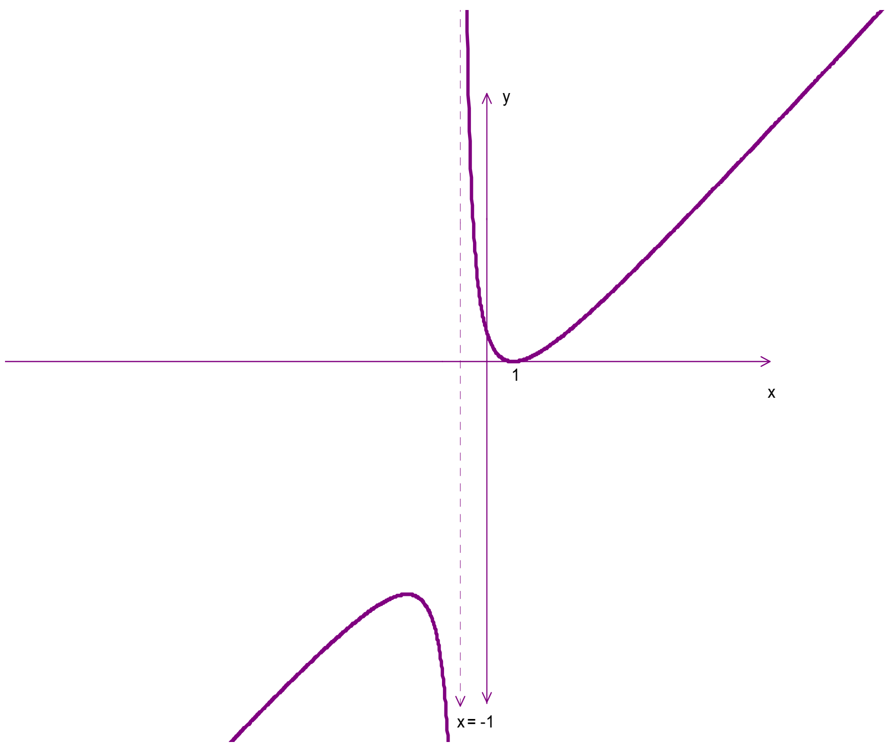
(c) Find the foot of the perpendicular from A to Π_2. **[4]**

10 The diagram shows a sketch of $y=\dfrac{(x-a)^2}{x+b}$. The curve has a vertical asymptote $x=-1$ and a minimum point at $(1,0)$.

(i) Find the values of a and b. Hence obtain the equation of the oblique asymptote. **[4]**

(ii) Without the use of graphic calculator, find the coordinates of the maximum point and deduce the set of values of k for which $\dfrac{(x-a)^2}{x+b}=k$ has no real solutions. **[3]**

(iii) The curve is translated p units along y-axis where p is positive. The resulting curve intersects the line $y=-6$ at exactly one point. Show that the equation of the resulting curve is $y=\dfrac{x^2+c}{x+1}$, where c is to be determined. Hence, draw a sketch of $y=\dfrac{x^2+c}{1-x}$, stating clearly the stationary points and equations of asymptotes. **[4]**



11 **(a)** By means of the substitution $y = vx^2$, show that the differential equation

$x\frac{\mathrm{d}y}{\mathrm{d}x} = 2y + x^2 \ln x$ can be reduced to $x\frac{\mathrm{d}v}{\mathrm{d}x} = \ln x$.

Hence obtain the general solution of the differential equation $x\frac{\mathrm{d}y}{\mathrm{d}x} = 2y + x^2 \ln x$, expressing y in terms of x. **[5]**

(b) In a laboratory, 1000 biscuits were exposed to particular bacteria that will cause the biscuits to become mouldy. After t days, there were x mouldy biscuits. It is assumed that the rate of increase of the number of mouldy biscuits is proportional to the product of the number of mouldy biscuits and the number of unaffected biscuits at any time t. Write down a differential equation involving x and t.

Given that $x = 1$ when $t = 0$, show that the solution of the differential equation is

$\frac{x}{1000 - x} = \frac{1}{999}e^{1000kt}$ where k is a constant. **[4]**

Given also that $x = 100$ when $t = 10$. How many more days does it take to have at least 500 mouldy biscuits? **[3]**

12 **(a)** Solve the equation $z^3 = \mathrm{i}$, giving your answers exactly in the form of $re^{\mathrm{i}\theta}$. Hence or otherwise, find the possible values of α in the interval $-\pi < \alpha \le \pi$ given that $w = e^{\mathrm{i}\alpha}$ and $w^* = \mathrm{i}w^2$ **[5]**

(b) Complex numbers z_1 and z_2 are given by

$$z_1 = 2\left(\cos\frac{\pi}{8} + \mathrm{i}\sin\frac{\pi}{8}\right), \qquad z_2 = 4\sqrt{2}\left(\cos\frac{5\pi}{8} + \mathrm{i}\sin\frac{5\pi}{8}\right)$$

Illustrate on an Argand diagram Z_1 and Z_2, the points representing the complex numbers z_1 and z_2 respectively and hence show that $|z_2 - z_1| = 6$. **[3]**

On the same diagram, sketch the loci

(i) $|z - z_1| = 6$, **[2]**

(ii) $\arg(z - z_1) = \frac{5\pi}{8}$. **[2]**

The complex number w is represented by the point of intersection of the loci in part (i) and (ii). Find the modulus and the argument of w. **[3]**